



Paper ID : 250519

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Subject Code: RCAI603

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MCA-INT
(SEM VI) THEORY EXAMINATION 2024-25
ARTIFICIAL INTELLIGENCE

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A**1. Attempt all questions in brief.****02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define Artificial Intelligence and mention two of its applications.	1	K1
b.	Define heuristic function with an example.	1	K1
c.	Define rational agent. How does it differ from a simple reflex agent?	1	K2
d.	What are the limitations of Depth-First Search?	2	K2
e.	What is the difference between propositional logic and first-order logic?	3	K2
f.	Define probabilistic reasoning.	3	K1
g.	List two advantages of using Naive Bayes in classification.	4	K2

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

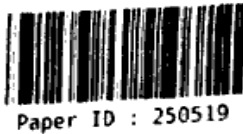
Q no.	Question	CO	Level
a.	Explain the structure and types of intelligent agents with diagrams and examples.	1	K3
b.	Describe A* search algorithm with an example. Highlight its optimality.	2	K3
c.	Explain backward chaining in first-order logic with suitable example.	3	K4
d.	Describe supervised and unsupervised learning with suitable examples.	4	K3
e.	Explain Hidden Markov Models (HMM) and their application in NLP or vision.	3	K4

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain applications of AI in real life focusing on robotics and expert systems.	1	K3
b.	Discuss the role of computer vision and NLP in intelligent systems.	1	K3

4. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Describe Hill Climbing and Simulated Annealing with comparisons.	2	K3
b.	Explain the Minimax algorithm and its improvement using alpha-beta pruning.	2	K4



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TIME: 3 HRS**M.MARKS: 70****5. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	What is unification in logic? How does it support inference in AI systems?	3	K3
b.	Suppose we are given: • $P(\text{Fever}) = 0.3$ • $P(\text{Malaria}) = 0.1$ • $P(\text{Fever} \text{Malaria}) = 0.8$ • $P(\text{Malaria} \text{Fever}) = ?$ Use Bayes' Theorem to compute the posterior probability $P(\text{Malaria} \text{Fever})$.	3	K4

6. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	You are given 4 data points in 1D space: {1, 2, 8, 9} Using $K = 2$, perform one iteration of the K-Means Clustering algorithm. Start with initial centroids: 1 and 8. Assign clusters and recompute centroids.	4	K4
b.	Explain reinforcement learning with an example from game playing or robotics.	4	K4

7. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Describe the working of the K-means algorithm and its limitations.	5	K3
b.	What is Linear Discriminant Analysis (LDA)? How does it differ from PCA?	5	K4